RAJALAKSHMI ENGINEERING COLLEGE

An Autonomous Institution, Affiliated to Anna University

Rajalakshmi Nagar, Thandalam – 602 105

Web Technology And Mobile Application

Experiment 9

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By

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AIM:

To develop a simple calculator Android app using Android Studio to perform basic arithmetic operations.

ALGORITHM:

- Create a new project in Android Studio.
- Design the layout using TextView and GridLayout with Buttons in XML.
- Define button styles in the resources file.
- Initialize UI components in MainActivity.java.
- Implement listeners for numeric buttons to take input.
- Implement listeners for operator buttons (+, −, ×, ÷).
- Write logic for the equal button to perform calculations.
- Handle special cases like division by zero.
- Add clear button functionality to reset values.
- Run the application and test all operations.

CODE:

XML:

```xml
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:background="#000"
    android:padding="16dp">

    <TextView
        android:id="@+id/resultText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="0"
        android:textSize="32sp"
        android:gravity="end"
        android:padding="16dp"
        android:background="#EEE"/>

    <GridLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:rowCount="5"
        android:columnCount="4"
        android:layout_marginTop="16dp"
        android:padding="8dp">
```

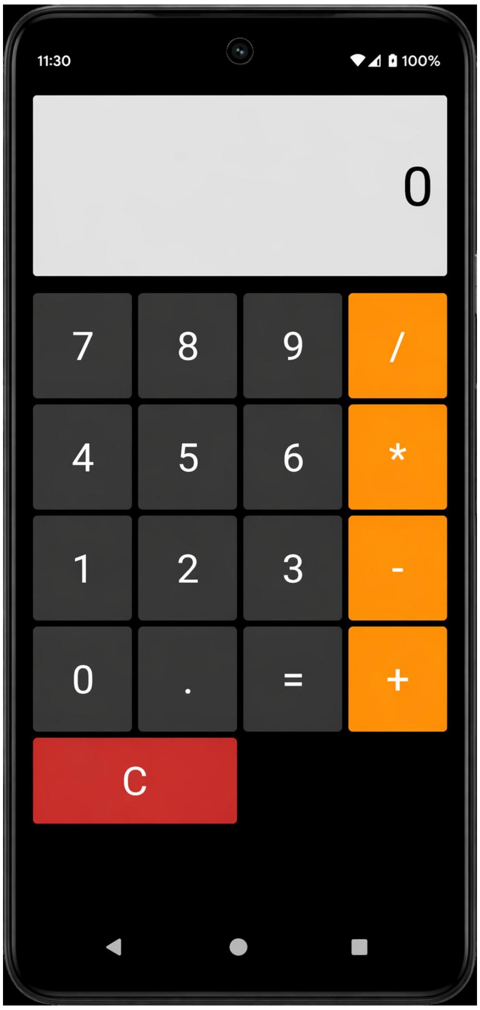
```xml
        <!-- Row 1 -->
        <Button android:id="@+id/button7" android:text="7" style="@style/CalculatorButton"/>
        <Button android:id="@+id/button8" android:text="8" style="@style/CalculatorButton"/>
        <Button android:id="@+id/button9" android:text="9" style="@style/CalculatorButton"/>
        <Button android:id="@+id/buttonDivide" android:text="/" style="@style/CalculatorButton"/>

        <!-- Row 2 -->
        <Button android:id="@+id/button4" android:text="4" style="@style/CalculatorButton"/>
        <Button android:id="@+id/button5" android:text="5" style="@style/CalculatorButton"/>
        <Button android:id="@+id/button6" android:text="6" style="@style/CalculatorButton"/>
        <Button android:id="@+id/buttonMultiply" android:text="*" style="@style/CalculatorButton"/>

        <!-- Row 3 -->
        <Button android:id="@+id/button1" android:text="1" style="@style/CalculatorButton"/>
        <Button android:id="@+id/button2" android:text="2" style="@style/CalculatorButton"/>
        <Button android:id="@+id/button3" android:text="3" style="@style/CalculatorButton"/>
        <Button android:id="@+id/buttonSubtract" android:text="-" style="@style/CalculatorButton"/>

        <!-- Row 4 -->
        <Button android:id="@+id/button0" android:text="0" style="@style/CalculatorButton"/>
        <Button android:id="@+id/buttonDot" android:text="." style="@style/CalculatorButton"/>
        <Button android:id="@+id/buttonEqual" android:text="=" style="@style/CalculatorButton"/>
        <Button android:id="@+id/buttonAdd" android:text="+" style="@style/CalculatorButton"/>

        <!-- Row 5 -->
        <Button
            android:id="@+id/buttonClear"
            android:text="C"
            style="@style/CalculatorButton"/>

    </GridLayout>

</LinearLayout>
```

MainActivity.java:

```java
package com.example.calculator;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    private TextView resultText;
    private String currentInput = "", operator = "";
    private double firstNumber = 0;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
```

```java
        resultText = findViewById(R.id.resultText);

        int[] numbers = {R.id.button0, R.id.button1, R.id.button2, R.id.button3,
                R.id.button4, R.id.button5, R.id.button6,
                R.id.button7, R.id.button8, R.id.button9, R.id.buttonDot};

        for (int id : numbers) {
            findViewById(id).setOnClickListener(v -> {
                currentInput += ((Button) v).getText();
                resultText.setText(currentInput);
            });
        }

        int[] operators = {R.id.buttonAdd, R.id.buttonSubtract,
                R.id.buttonMultiply, R.id.buttonDivide};

        for (int id : operators) {
            findViewById(id).setOnClickListener(v -> {
                if (!currentInput.isEmpty()) {
                    firstNumber = Double.parseDouble(currentInput);
                    operator = ((Button) v).getText().toString();
                    currentInput = "";
                }
            });
        }

        findViewById(R.id.buttonEqual).setOnClickListener(v -> {
            if (!currentInput.isEmpty()) {
                double secondNumber = Double.parseDouble(currentInput);
                double result = 0;

                switch (operator) {
                    case "+": result = firstNumber + secondNumber; break;
                    case "-": result = firstNumber - secondNumber; break;
                    case "*": result = firstNumber * secondNumber; break;
                    case "/": result = secondNumber != 0 ? firstNumber / secondNumber : 0; break;
                }

                resultText.setText(String.valueOf(result));
                currentInput = "";
            }
        });

        findViewById(R.id.buttonClear).setOnClickListener(v -> {
            currentInput = operator = "";
            firstNumber = 0;
            resultText.setText("0");
        });
    }
}
```

OUTPUT:



RESULT:

The calculator app runs successfully and performs addition, subtraction, multiplication, and division correctly.